

Mapping the Consumption Gap

Planned Data Center Capacity versus Electricity Supply in California

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Defense deck. Data frozen September 22, 2026

The question, borrowed from Finland

- ▶ Finland (Manner, IRTF SUSTAIN, July 2026): about 285 MW of data centers operating, 7,777 MW in the public investment registry.
- ▶ California, December 2025: 23,277 MW of energization requests reported to the CEC, 20,677 MW in the state: 5,086 MW signed, 6,987 MW applications, 8,604 MW inquiries.
- ▶ Existing California data center load: about 1,000 MW, 7.0 to 9.3 TWh a year.
- ▶ The CEC discounts requests with confidence levels, a utilization factor and a ramp. Are the discounts right, can supply keep up, what can flexibility do, and how does the count compare with other regimes?

Sources: manner2026; cec_assembly_hearing_2026_01_28; cec_dc_methodology_memo_2026; chapter 1 estimates (ch1_dc_load_estimates.csv).

Research questions and the method

- ▶ **RQ1** Pipeline by stage, absolute and relative to the grid.
- ▶ **RQ2** The 2030 gap, raw and probability-weighted, in TWh and net-peak MW.
- ▶ **RQ3** Six counting regimes compared; California's number under each rule.
- ▶ **RQ4** Headroom from load flexibility as a share of the gap.
- ▶ An independent replication and audit of the CEC method.
- ▶ Every raw file in a manifest (URL, access date, SHA-256); one command per chapter; every figure stamped with its files and manifest ids.
- ▶ Data frozen 2026-09-22; repository tagged; processed outputs exported with the report.

Chapter 1: the grid as it stands

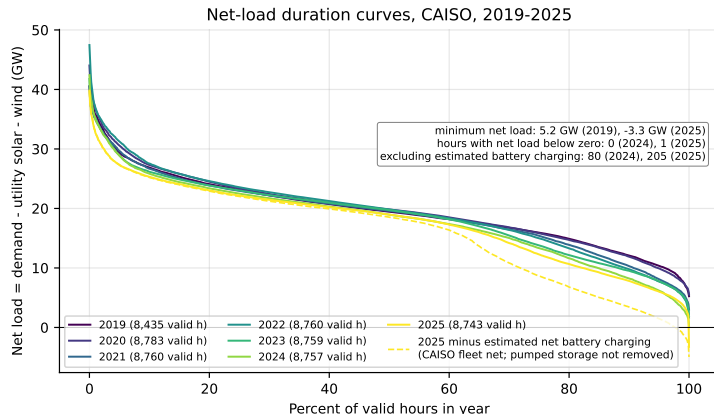


fig1_02. Data: ch1_net_load_duration_curves.csv; ch1_ciso_annual_summary.csv; ciso_hourly_2019_2025_clean.parquet. Sources (manifest ids): eia930 balance files; caiso_outlook fuelsource (Batteries). Code: ch1_baseline.add_storage_adjusted, duration_curves; run_chapter1 step 1. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

- ▶ CAISO demand 2019 to 2025 from EIA-930, cleaned against CAISO's own series; record peak 52,061 MW (2022), 43,860 MW in 2025.
- ▶ Net load after solar and wind now falls below zero for hundreds of hours a year; the evening net peak is what the fleet is planned for.

Chapter 1: prices, carbon and existing data center use

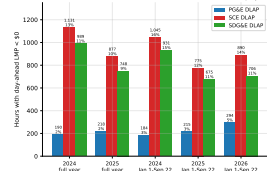
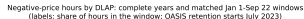


Fig. 56. Data: ch1_negative_price_hours, windows.csv; ch1_negative_price_hours_by_month.csv; ch1_price_stats.csv. Sources: Inlandnet (ch1_price_stats.csv), log monthly Res. Code: ch1_baseline_negative_price_hours_by_month_price_stats_nin_chapter1_step2. Data from 2020-08-22. Data from Inlandnet.csv, Inlandnet.csv, access stats and 598,294 for each id.

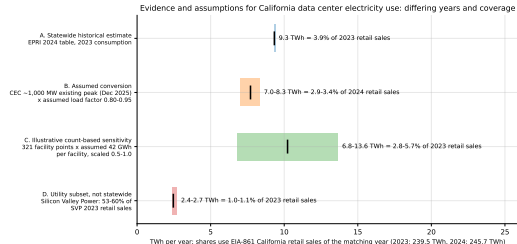
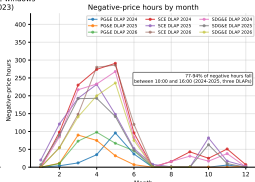
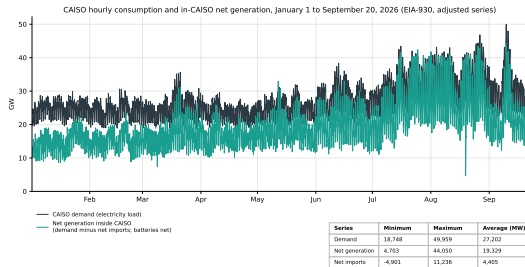


Fig. 88. Data: chi_d, lead estimates csc_chi, c retail sales eis003 csc_chi2, svp fact sheets csc_chi, kollar grady points california flag csc_chi bottom up inputs pom Sources (manifest kb):
 april_governing_intelligence_2024 csc_chi_d methodology memo_2026 eis003_2019_2024 svp fact sheet 2017_2022 kollar grady 2025 jeroski opex data centers core_chi baseline d lead estimates bottom up estimates
 c retail sales eis003 svp fact sheet energy kollar grady nrc chapter 30 data 4 Data from 2014-09-22 data4value/manifest/csc_chi01d, access date and SHA-256 for each id.

- ▶ Negative day-ahead prices in 877 hours at the SCE load-aggregation point in 2025 (1,131 in 2024), 78 percent between 10 and 16 h.
- ▶ Existing data center use, four ways: a statewide estimate, an assumed conversion of the CEC's 1,000 MW, a count-based sensitivity and the Silicon Valley Power cluster. The spread is the finding.

Chapter 1: the same five charts as the Finnish study, for CAISO



Figt. 0% Data: `ch1_h0daily_2020_workshop.csv`; `ch1_workshop_stats.csv`; `cabo_ch2_intensity_hourly_2020.parquet`; `cabo_kutrix_hourly_2020.parquet`. Sources (`intensity_ch1_eia930_balance_2020_jan_jun_2020_jul_dec`); `cabo_nutbok_demand` and `fuelsource_2020` (artifical filter). Code: `ch1_baseline_load_eia930_cbo_clean_demand_against_cabo_hourly_stats_run_chapter1` step 5. Data frozen 2020-09-22; `datahiveintartifical.csv` holds URL, access date and SHA-256 for each file.

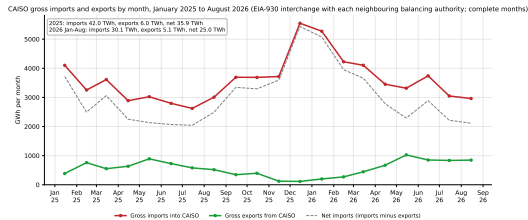


Fig. 12. Data: *ch1_imports_exports_monthly*; *ch1_interchange_gross_2025_2026_pumped*. Sources (manifest id): *ch1930_interchange_2025_jan_jun* - 2026_jul_dec; *ch1930_balance_2025_jan_jun* - 2026_jul_dec (not check). Code: *ch1_balance_load* *ch1930_interchange* *ch1_monthly_imports_exports* *ch1_charts* *ch1* v. 5. Data spans 2024.05.15 - 2024.06.01. Data is manifest id: *ch1930*. Access data and 500,000 for each id.

- ▶ January to September 2026: demand 18,748 to 49,959 MW, mean 27,202 MW; generation inside CAISO covered a mean 19,329 MW and net imports 4,405 MW.
- ▶ 2025: 42.0 TWh of gross imports against 6.0 TWh of exports; wind ran at 26 percent and solar at 25 percent of nameplate in 2026; the consumption CO₂ factor averaged 163 g/kWh against Finland's 29 to 31.

Chapter 2: construction broke upward after ChatGPT

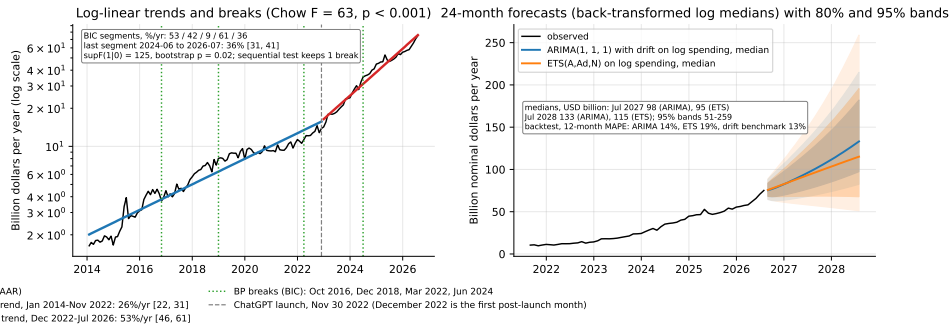


fig2_02 Data: ch2_census_break_summary.json; ch2_census_bai_perron.csv; ch2_census_segment_growth.csv; ch2_census_forecast_arima.csv; ch2_census_forecast_ets.csv; ch2_census_forecast_backtest.csv. Sources (manifest ids): census_c30_privsatime. Code: ch2_growth.chow_test, bai_perron, sequential_supf, arima_forecast, ets_forecast, backtest_forecasts; run_chapter2 step 2. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

- Census data center construction spending: a Chow test at December 2022 and Bai-Perron breaks located by the data agree; ARIMA and ETS bands and a rolling-origin backtest.
- California has no state construction series; proxies are QCEW employment, CBRE Silicon Valley MW and the Epoch AI hub.

RQ1: the pipeline against the grid

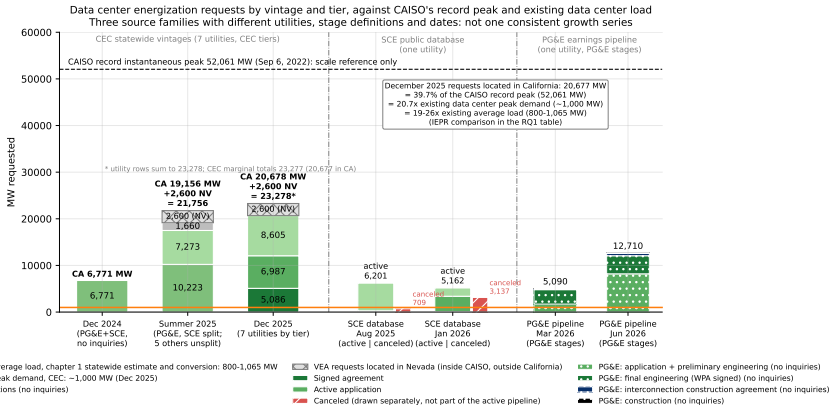
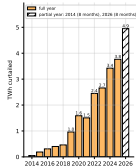
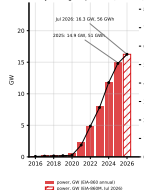
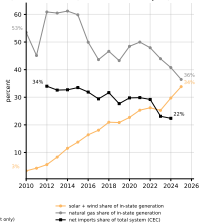
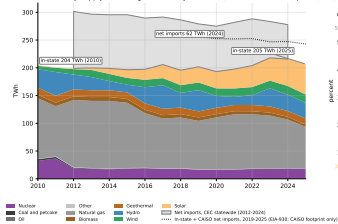


fig2_04. Data: ch2_tier_vintages.csv; ch2_rq1_denominators.json; ch1_dc_load_estimates.csv. Sources (manifest ids): cec_dc_forecast_2024iepr; cec_prelim_dc_forecast_2025; cec_dc_methodology_memo_2026; cec_assembly_hearing_2026_01_28; cec_tn266008; cec_tn268459; cec_tn272026; cec_tn272065; cec_tn272807; caiso_key_statistics_2026_08. Code: ch2_growth_tier_vintages; run_chapter2_steps_4_and_5. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

- ▶ 20,677 MW in California: 40 percent of the CAISO record peak, twenty times the existing data center peak, five times the IEPR's 2025 to 2030 managed peak growth.
- ▶ VEA's 2,600 MW is in Nevada; SCE's cancellations grew from 709 to 3,137 MW between vintages.

Chapter 3: supply history and the fleet



- ▶ In-state generation 204 TWh (2010) and 205 TWh (2025); gas 53 to 36 percent, solar and wind 3 to 34 percent; imports 103 to 62 TWh (2012 to 2024).
- ▶ Capacity 73 to 107 GW; batteries 15 GW and 51 GWh; curtailment 3.8 TWh in 2025 and 4.9 TWh in January to August 2026.

Chapter 3: what gets built, from California's own record

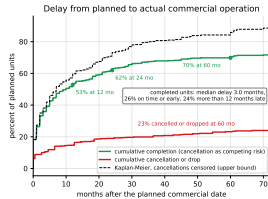
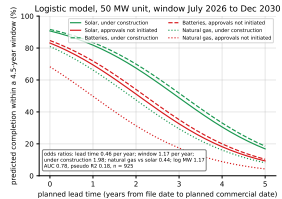
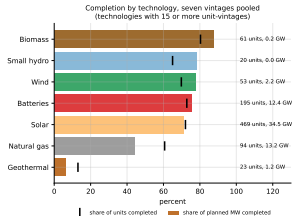
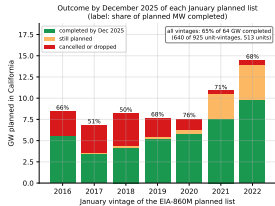


Fig. S4. Data: ch3_vintage_outcomes.csv; ch3_realization_by_vintage.csv; ch3_realization_by_technology.csv; ch3_logit_coefficients.csv; ch3_logit_summary.csv; ch3_delay_km.csv; ch3_delay_of.csv. Sources: manifest data: eia860m 2016_03_2022_01; eia860m 2016_02_2022_02. Code: ch3_supplyeia860m_vintage_outcomes, fit_completion_logit, km_delay, run_chapter2 step 2. Data frozen 2026-09-22; dataraw/manifest.csv holds URL, access date and SHA-256 for each id.

- ▶ 925 unit-vintages of planned generators, January 2016 to 2022, followed to December 2025: 65 percent of megawatts operating; gas 44 percent, geothermal 6 percent.
- ▶ Logistic completion model (lead time, window, status, technology, size) and delay curves with cancellation as a competing risk.

Chapter 3: probability-weighted supply in 2030

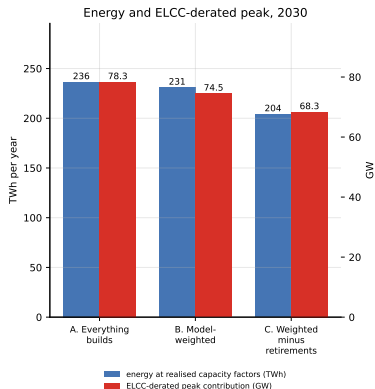
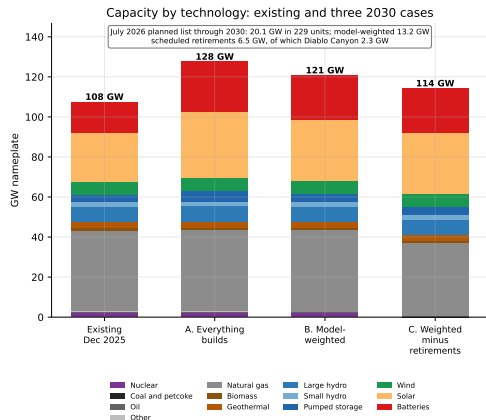


fig3_05. Data: ch3_cases_2030.csv; ch3_cases_2030_summary.json; ch3_planned_current_weighted.csv; ch3_existing_capacity_2025.csv; ch3_capacity_factors.csv; ch3_elcc_values.csv. Sources (manifest ids): eia860m_2026_07; eia860m_2025_12; cpuc_e3_astrape_incremental_elcc_2023. Code: ch3_supply/current_planned, cases_2030, capacity_factors, elcc_table; run_chapter3 step 4. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

- Everything builds: 128 GW, 236 TWh, 78 GW of ELCC. Model-weighted: 121 GW, 231 TWh, 74 GW. Minus 6.5 GW of retirements including Diablo Canyon: 114 GW, 204 TWh, 68 GW.

Chapter 4: the same requests under four counting rules

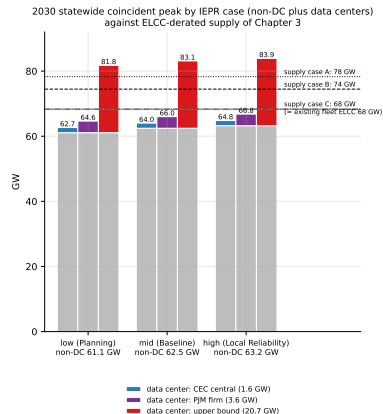
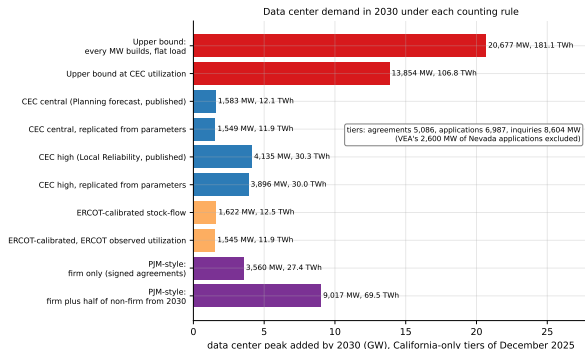


fig4_01. Data: ch4_demand_cases_2030.csv; ch4_demand_totals_2030.csv; ch4_cec_replication.csv; ch4_cec_ramp_profile.csv; ch4_ca_tiers.csv; ch4_iepr_growth_summary.csv; ch4_ercot_chain.csv; ch3_cases_2030_summary.json. Sources (manifest ids): cec_assembly_hearing_2026_01_28; cec_dc_methodology_memo_2026; cec_in268722_268727 (CED 2025 forms); cec_in268124; cec_in268824; ERCOT decks and monthlies (see fig4_05); pjm_lar_summary_2025_11_24. Code: ch4_gap.ca_tiers, cec_parameters, cec_forecast_replication, ced2025_scenarios, iepr_growth_cases, ercot_chain, demand_cases_2030, demand_totals_2030; run_chapter4 steps 1-3. Data frozen 2026-09-22; data/manifest.csv holds URLs, access data and SHA-256 for each id.

- Upper bound 20,677 MW and 181 TWh; CEC central 1,583 MW (rebuilt from its parameters: 1,549); CEC high 4,135; ERCOT-calibrated 1,622; PJM firm-only 3,560.
- The CEC parameters with SVP exempt reproduce its 2040 endpoints (4,855 and 7,381 MW) to within 1 MW.

Data centers as a share of California's electricity

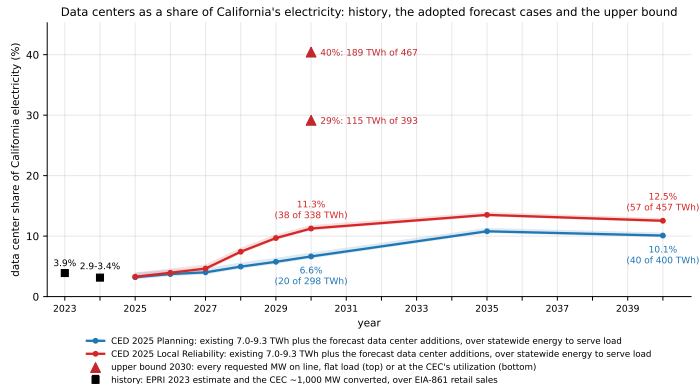
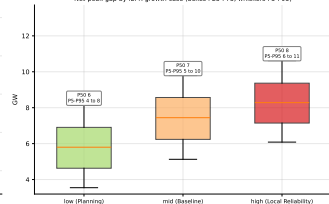
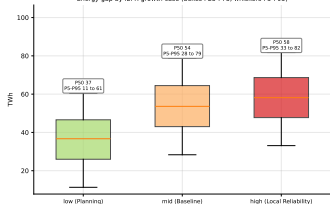
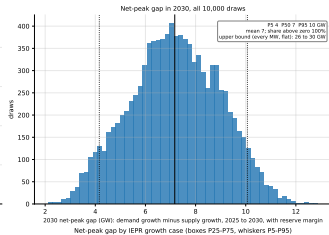
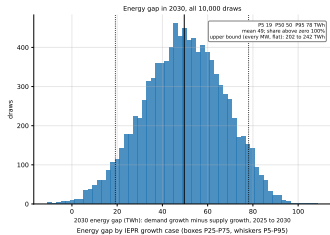


fig4_08. Data: ch4_dc_share_trajectory.csv; ch1_dc_load_estimates.csv; ch4_ced2025_data_center_energy.csv; ch4_ced2025_scenarios.csv; ch4_demand_cases_2030.csv. Sources (manifest ids): epri_powering_intelligence_2024; cec_dc_methodology_memo_2026; eia861_2019..2024; cec_tn268727; cec_tn268725; cec_tn268824; cec_tn268124; cec_assembly_hearing_2026_01_28. Code: run_chapter4 step 9. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

- 3.9 percent of retail sales in 2023; 6.6 percent in 2030 and 10.1 percent in 2040 under the adopted Planning forecast; 11.3 and 12.5 percent under Local Reliability; 40 percent if every request ran flat by 2030 (29 percent at the CEC's utilization). Oregon: 23 percent today, 31 to 32 percent by 2030.

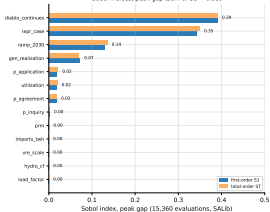
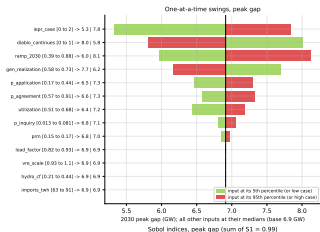
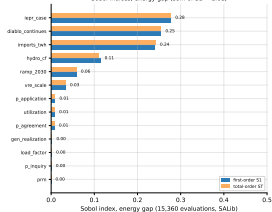
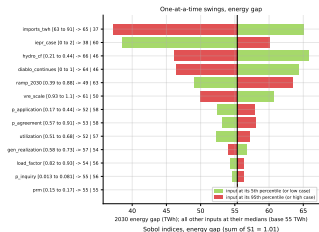
RQ2: the 2030 gap, 10,000 draws



Byd_02, Oskc, ch4_mrc, drwa, parqut, ch4_mrc_summary.csv, ch4_upper_bound_2030.csv, ch4_mrc_inputs.csv, ch4_mrc_key_numbers.json. Sources (manifest.html): chapter 3 outputs (eia900r_2026_07, eia900r_2025_12, cpuc, e3, estape, incremental_elec_2023, eia2033, cec, elec, energy_generation_page1, CEO_2025_ferms) as above. Code: ch4_ops, SocioRefModel, mrc_inputs, moreca, carls, evaluate, aap, summarise, upper_bound_mrc; run chapter4_step4. Data from 2014-09-22: data.gov/manifest.csv holds URL, access date and SHA-256 for each id.

- Energy gap 50 TWh at the median (19 to 78); net-peak gap 7.2 GW (4.2 to 10.1). Data centers: 18 TWh and 2.3 GW of it.
- Upper bound: 202 to 242 TWh and 25.5 to 30.2 GW, four to five times the weighted 95th percentile. Gas at a 0.40 capacity factor would close the median energy gap.

What drives the answer

[illegible]

- Energy gap: imports, the IEPR case, hydro and Diablo Canyon; the three tier probabilities carry about 2 percent of the energy gap's variance and 4 percent of the peak gap's.
- Peak gap: Diablo Canyon (total index 0.39), the IEPR case (0.35), the data center ramp (0.14), generation completion (0.07).

RQ4: headroom from flexibility



- ▶ Duke's method on CAISO load 2019 to 2025: 3.1, 3.8, 4.6 and 8.0 GW at 0.25, 0.5, 1 and 5 percent curtailment (Duke: 4.2, 5.0, 5.9); 98 percent of it in November to February.
- ▶ 3.8 GW is 2.4 times the CEC central data center addition, 53 percent of the median peak gap, 13 percent of the upper-bound gap.

Flexibility also cuts emissions

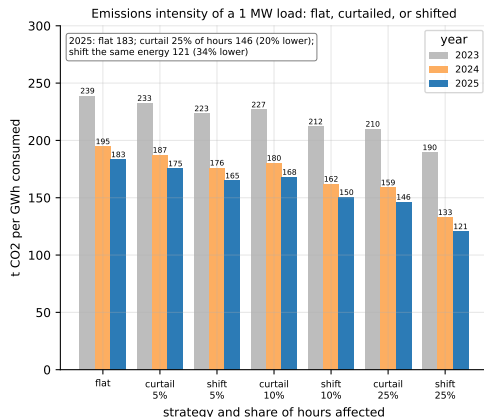
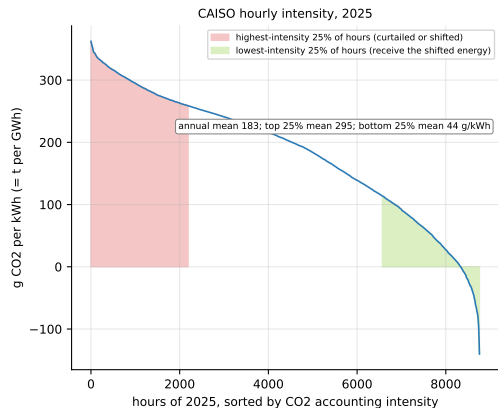


fig4_06. Data: ch4_emissions_flat_vs_flexible.csv; caiso_co2_intensity_hourly.parquet. Sources (manifest ids): caiso_outlook co2 and demand daily files. Code: ch4_gap.flat_vs_flexible_emissions; run_chapter4 step 8. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

- A flat load carried 183 t of CO₂ per GWh in 2025; curtailing the highest-intensity quarter of hours cuts that by 20 percent, shifting the energy into the lowest-intensity quarter by 34 percent.

Contribution: a replication and audit, not a new forecast

- ▶ The CEC's published parameters rebuild its forecast: exact at the 2040 endpoints, within 2 to 6 percent in 2030 once the SVP exemption is applied.
- ▶ The supply side is discounted with the same discipline, from California's own completion record.
- ▶ The confidence levels are tested against ERCOT's realized transition rates and PJM's firm rule; the gap is a distribution with a stated upper bound and named drivers.
- ▶ Headroom replicated on CAISO load and expressed as a share of the gap; six regimes scored on one rubric.
- ▶ Everything reproducible: manifest, one command per chapter, stamped figures, frozen data, tagged repository, exported outputs.

Policy implications

- ▶ **CEC:** publish requests at project level with size, requested year and status history (as SCE does); report monthly by phase (as ERCOT does); publish firm and non-firm components separately (as PJM does); state the SVP exemption and the Nevada request where totals appear.
- ▶ **CPUC and CAISO:** a flexible interconnection service is worth 3 to 5 GW of headroom, mostly winter evenings, plus the midday surplus energy; FERC's June 2026 orders ask for exactly that.
- ▶ **Resource planning:** weight planned generation by completion probability, updated yearly from EIA-860M.
- ▶ **Legislature:** Diablo Canyon moves the gap more than any data center assumption (18 TWh, 2.2 GW); licences are in hand, state law stops at 2030.
- ▶ **EIA:** extend the pilot survey to California.

Limits

- ▶ Boundaries: CAISO hourly data against statewide generation and forecast; imports' peak contribution outside the absolute balance.
- ▶ ERCOT hazards measured during a surge; several ERCOT and PJM values read from chart images.
- ▶ ELCC from the CPUC's 2023 study; reserve margin an assumed range; Diablo Canyon a coin flip with the licence facts stated.
- ▶ Data center load flat at the peak; 2030 energy annualized at the end-2030 level.
- ▶ No project-level California request data outside SCE; the CEC's utility ramp schedules are not public.

Conclusion

- ▶ **RQ1** 20,677 MW in three tiers: 40 percent of the record peak, twenty times existing data center load.
- ▶ **RQ2** 2030 gap 50 TWh and 7.2 GW at the median (data centers 18 TWh, 2.3 GW); upper bound 202 to 242 TWh, 25 to 30 GW.
- ▶ **RQ3** Same requests: 20,677 / 5,086 / about 2,400 / about 1,000 MW under the four counting rules; regimes differ most in timeliness, verification and access.
- ▶ **RQ4** 3.8 GW of headroom at 0.5 percent curtailment: twice the weighted data center load, half the weighted peak gap, under a fifth of the upper bound.
- ▶ The method is the result: a forecast anyone can rebuild, a supply side discounted the same way, a gap with its uncertainty and drivers named.

Backup: reproducibility

- ▶ Repository: `src/` (`ch1_baseline`, `ch2_growth`, `ch3_supply`, `ch4_gap`, `ch4_regimes`, `provenance`), `scripts/run_chapter1..4.py`, `make ch1..ch4`, `make provenance`, `make freeze`, `make release`.
- ▶ Raw store: `data/raw/<group>/<access date>/`, immutable; `manifest.csv` with URL, access date, SHA-256; frozen copy `manifest_frozen_2026-09-22.csv`.
- ▶ Every figure stamped with its processed files, manifest ids and code; Appendix A repeats them as tables; `docs/DATA_NOTES.md` records every limit found.
- ▶ Standalone bundle on the Desktop reproduces every output byte for byte in its own environment.